ChatGPT

OFFOLabs EV Fit Questionnaire and Recommendation Engine

Executive summary

You already have the right core idea: start with charging context, driving demand, climate, and "hard days." To reliably recommend a *perfect-fit* EV, OFFOLabs needs to (a) separate **hard constraints** (charging feasibility, budget, required range buffer) from (b) **preference ranking** (utility, tech, comfort, ownership friction). It also needs transparent sourcing and user-facing explanations tied to authoritative datasets.

This handoff delivers:

- A **final progressive questionnaire** with exact UI copy, input types, defaults, and branching rules for **Quick Fit (2–4 minutes)** and **Deep Fit (8–12 minutes)**.
- A transparent **scoring model** with constraint filtering + weighted preference scoring, including buffer multipliers for climate, towing/load, parking exposure, and minimum-comfort SOC (state of charge).
- **Worked numeric examples for three archetypes**, using a small illustrative vehicle set (not real vehicle claims) to show how the engine ranks options.
- Recommendation explanation **templates** that list inputs, sources, timestamps, and limitations (EPA range/efficiency, NHTSA VIN decode/recalls, AFDC/NREL charger density, EIA electricity price).
- A branching **Mermaid flowchart**, an analytics **event mapping table**, privacy/paywall copy snippets, and a short **A/B test plan**.

Authoritative sources used throughout include: U.S. Environmental Protection Agency [1] fuel economy and EV testing methodology (official range/MPGe backing), National Highway Traffic Safety Administration [2] vPIC VIN decoding and recalls datasets, Alternative Fuels Data Center [3] and National Renewable Energy Laboratory [4] station data/APIs, American Automobile Association [5] range and cargo studies, and U.S. Energy Information Administration [6] electricity price datasets. [7]

Assumptions: OFFOLabs has (or will build) an internal vehicle catalog keyed by make/model/year/trim with EPA label range and efficiency; fast-charging characteristics may come from OEM specs or partners and are marked as "non-EPA" where used. Live API keys and data refresh schedules are unspecified and are treated as implementation parameters.

Data sources and design principles

Authoritative data foundation

OFFOLabs can ground its recommendation engine in public, defensible data sources:

- **EPA / FuelEconomy.gov**: downloadable fuel economy data used for official label values. FuelEconomy.gov explicitly states that fuel economy data are the result of testing at EPA's NVFEL and manufacturers with EPA oversight. [8]
- **EPA test methodology**: EPA describes the 5-cycle approach including hot (95°F) and cold (20°F) tests with HVAC settings, which supports explaining why climate buffers exist. [9]
- **NHTSA vPIC VIN decode**: the vPIC API documentation recommends sending model year for more accurate decoding and supports partial VIN decode for flat format endpoints. [10]
- **NHTSA recalls**: NHTSA provides recall lookup and maintains "datasets and APIs" including recalls data. [11]
- **Public charging density**: NREL's Alternative Fuel Stations API powers the AFDC station locator and supports "nearest" and "nearby route" endpoints. [12]
- **Charging speed expectations**: DOT's charging toolkit describes Level 1 vs Level 2 typical charge times; AFDC guidance provides general charging context. [13]
- **Range buffer evidence**: AAA's EV range testing reports HVAC/cold impacts and provides quantitative range reduction context (e.g., 41% reduction with HVAC at 20°F vs 75°F). [14]
- **Load/payload effects**: AAA shows significant range reduction when hauling heavy cargo (example reduction ~24.5% in a payload test case). [15]
- **Electricity prices**: EIA provides electricity sales/revenue/price tables by sector and state, supporting "estimated cost per mile" when users don't know their rate. [16]

UX principles for a "perfect fit" tool

- **Progressive disclosure**: ask only what you need to compute an initial shortlist, then deepen.
- **No surprise outcomes**: show before/after impact: "This answer increases recommended range by X%" and surface pricing gates early (if any).
- **Explain buffers**: climate and load multipliers must be defensible and interpretable; cite AAA and EPA methodology. [17]
- **Privacy-first inputs**: ZIP and VIN should be requested with clear purpose and retention controls; NHTSA has described VIN as PII in vPIC materials. [18]

Progressive questionnaire flows with exact UI copy and branching

Quick Fit flow

Goal: produce a credible shortlist fast, even when users are uncertain. Expected completion: 2–4 minutes.

Screen 0: Welcome - **Title:** "Find your best-fit EV in minutes" - **Body:** "Answer a few questions and we'll rank EVs based on range needs, charging reality, budget, and your preferences. You'll see *why* each vehicle matches and what assumptions we used." - **Primary button:** "Start Quick Fit" - **Secondary link:** "Go straight to Deep Fit (more accurate)"

Step 1: Charging anchor - **Question (UI copy):** “Where will you charge most weeks?” - **Helper text:** “Pick the place you’ll reliably charge on most weeks. This sets your ‘charging reality’.” - **Input type:** Single-select cards (radio) - **Options (default order):** - “Home (garage / driveway)” - “Work (workplace charger)” - “Public charging (fast chargers + public Level 2)” - “Not sure yet” - **Branching rules:** - If **Home** → show Step 1a (home access mini-check) - If **Work** → show Step 1b (work access mini-check) - If **Public** → show Step 1c (public reliance mini-check) - If **Not sure** → skip mini-check and continue

Step 1a: Home access mini-check - **Question:** “Do you have a dedicated parking spot where you can plug in overnight?” - **Helper:** “Even a standard outlet can work for some drivers; Level 2 is faster.” - **Input:** Single-select - **Options:** “Yes” / “Sometimes” / “No” - **Branching:** If “No” → later in Deep Fit, prioritize public-dependent flow; Quick Fit continues.

Step 1b: Work access mini-check - **Question:** “How many days per week can you charge at work?” - **Input:** Stepper (0–5), default 3 - **Branching:** If 0–1 → treat as partial; Quick Fit continues.

Step 1c: Public reliance mini-check - **Question:** “How do you feel about waiting 20–40 minutes at a fast charger?” - **Input:** Single-select - **Options:** “Fine” / “Sometimes OK” / “I’d rather avoid that” - **Branching:** influences “public friction” and fast-charge weighting.

Step 2: Typical driving - **Question:** “About how many miles do you drive in a typical week?” - **Input:** Numeric with unit - **Default placeholder:** “e.g., 200” - **Validation:** 0–2,000 miles/week - **Helper:** “We’ll follow up with your ‘hard day’ next.”

Step 3: Hard day - **Question:** “What’s your longest driving day in the last month?” - **Input:** Numeric, miles - **Default placeholder:** “e.g., 140” - **Helper:** “This matters more than weekly miles for EV satisfaction.”

Step 4: Longer-trip frequency - **Question:** “How often do you have a longer-than-usual driving day?” - **Helper:** “Think: road trips, long meetings, airport runs, family days.” - **Input:** Single-select - **Options:** - “Weekly (about once a week)” - “Monthly (a few times a month)” - “Rarely (a few times a year)” - **Branching:** If Weekly or Monthly → later recommend stronger road-trip features.

Step 5: Budget - **Question:** “What’s your target purchase budget?” - **Helper:** “Use a max price to keep recommendations realistic.” - **Input:** Numeric ($) + optional toggle - **Defaults:** empty; quick buttons: “<$30k”, “$30–40k”, “$40–55k”, “$55k+” - **Toggle:** “Prefer monthly payment instead” - If toggled: ask “Estimated monthly payment target” (numeric, $/mo) and optionally “Lease” vs “Buy”

Step 6: Body style - **Question:** “What body style fits your life?” - **Input:** Single-select chips (allow “Any”) - **Options:** “Any”, “Sedan”, “Hatchback”, “Crossover”, “SUV”, “Truck / towing”, “Van / 3-row” - **Branching:** If “Truck / towing” → set towing/payload concerns and suggest Deep Fit towing module.

Step 7: Climate and location - **Question:** “Enter your ZIP code to estimate weather impact and nearby charging” - **Helper:** “We use your ZIP to estimate climate buffer and public charging density. You can skip.” - **Input:** ZIP (5 digits) + checkbox “Skip” - **If skip:** show manual climate question: - “What’s your climate like most of the year?” → Cold / Mild / Hot - **Branching:** ZIP also triggers AFDC charger density lookup (later explanation).

Quick Fit result screen - **Title:** “Your best-fit EV short list” - **Sections:** - “Top matches (why they fit)” - “Not recommended (why they don’t fit)” - “Your biggest constraint: [charging / range / budget]” - **CTA buttons:** “Refine with Deep Fit (recommended)” and “Compare a specific EV (VIN or listing)”

Deep Fit flow

Goal: improve accuracy by capturing charging feasibility, anxiety buffer, load, and ownership friction. Expected completion: 8–12 minutes depending on branching.

Deep Fit starts with a banner:
“Deep Fit improves accuracy by modeling charging speed, climate buffers, load/towing, and your comfort buffer.”

Module A: Charging reality

A1: Living situation - **Question:** “What best describes where you park overnight?” - **Input:** Single-select - **Options:** “Private driveway/garage”, “Assigned apartment/condo spot”, “Street/lot (no guaranteed spot)”, “Varies” - **Branching:** If not private → show A2b and A3.

A2a: Home charging setup (private parking) - **Question:** “What can you plug into at home?” - **Input:** Single-select - **Options:** - “Standard outlet (120V)” - “Level 2 charger already installed” - “Not sure / I’d need to check” - **Helper:** “Level 1 is slower; Level 2 is faster. We’ll estimate what you need.” [13]

A2b: Permission + installability (non-private parking) - **Question:** “Could you install a charger where you park?” - **Input:** Single-select - **Options:** “Yes”, “Maybe (need permission)”, “No” - **Follow-up if “Maybe”:** “Who would need to approve it?” → “Landlord”, “HOA/condo board”, “Employer”, “Other”

A3: Charging dwell time - **Question:** “How long is your EV typically parked and able to charge at your main charging spot?” - **Input:** Single-select - **Options:** “<4 hours”, “4–8 hours”, “8–12 hours”, “12+ hours” - **Helper:** “Dwell time determines how much range you can add per day.” (See Level 1/2 speed context.) [13]

A4: Public fallback - **Question:** “If you needed it, how often could you use public fast charging?” - **Input:** Single-select - **Options:** “Never”, “1–2×/month”, “1×/week”, “2+×/week” - **Branching:** If ≥1×/week → show A5.

A5: Public patience + friction - **Question:** “How do you feel about charging stops on road trips?” - **Input:** Single-select - **Options:** - “I’ll plan and stop—no big deal” - “I’ll do it, but I want fewer/shorter stops” - “I want to avoid fast charging as much as possible”

Module B: Driving patterns and range

B1: Typical day pattern - **Question:** “On a typical weekday, how many miles do you drive?” - **Input:** Numeric miles/day (default blank; helper “e.g., 24”)

B2: Highway share - **Question:** “Roughly what percent of your driving is highway?” - **Input:** Slider 0–100% (default 50%)

B3: Longest day cadence - **Question:** "Your longest driving day in a typical year is about…" - **Input:** Numeric miles (helper "e.g., 300") - **Branching:** If >250 → increase road-trip weighting.

B4: Destination charging - **Question:** "On long days, do you usually have charging at your destination?" - **Input:** Single-select - **Options:** "Yes, usually", "Sometimes", "No", "Not sure"

B5: Comfort buffer (range anxiety parameter) - **Question:** "What's the lowest battery you feel comfortable arriving with?" - **Input:** Single-select - **Options:** "10%", "20%", "30%", "40%" - **Helper:** "Higher comfort buffers mean you need more rated range."

Module C: Climate and storage

C1: ZIP-based climate - **Question:** "Confirm your ZIP code (for climate + charger density)" - **Input:** ZIP (prefill if entered in Quick Fit) + "Prefer not to share" - **If prefer not:** ask manual climate: Cold / Mild / Hot (same as Quick Fit)

C2: Outdoor exposure - **Question:** "Where is your EV parked most nights?" - **Input:** Single-select - **Options:** "Garage", "Driveway/covered", "Outdoors (exposed)", "Varies" - **Helper:** "Cold starts + cabin heat can reduce real-world range." [17]

C3: Winter severity (only if Cold climate) - **Question:** "How often do you drive in freezing temps (≤32°F)?" - **Input:** Single-select - **Options:** "Most weeks", "Sometimes", "Rarely" - **Helper:** "HVAC use at ~20°F can reduce range substantially in testing." [19]

Module D: Utility and hobbies

D1: Passengers - **Question:** "How many seats do you need most of the time?" - **Input:** Single-select - **Options:** "2", "4–5", "6–7", "8+"

D2: Cargo reality - **Question:** "What do you regularly haul?" - **Input:** Multi-select - **Options:** "Groceries", "Stroller/kids gear", "Pets", "Sports gear", "Tools", "Large items (IKEA/DIY)" - **Branching:** If tools/large items → increase cargo weighting.

D3: Towing and racks - **Question:** "Do you tow or carry heavy loads?" - **Input:** Single-select - **Options:** "No", "Occasionally", "Frequently" - **If Occ/Freq:** follow-up: - "Typical load or trailer weight?" → numeric lbs (optional)
- "Roof rack / hitch bike rack often?" → Yes/No
- **Helper:** "Heavy loads can meaningfully reduce range in testing." [15]

Module E: Budget and ownership

E1: New vs used - **Question:** "Are you shopping new, used, or either?" - **Input:** Single-select - **Options:** "New", "Used", "Either" - **Branching:** influences price_model and availability assumptions.

E2: Purchase plan - **Question:** "Do you plan to buy or lease?" - **Options:** "Buy", "Lease", "Not sure"

E3: Electricity cost - **Question:** "Do you know your electricity rate at home?" - **Input:** Single-select - **Options:** "Yes (I'll enter it)", "No (estimate for my state)", "I don't charge at home" - **If enter:** numeric $/kWh - **If estimate:** use EIA state residential average price table (show source in text). [20]

Module F: Tech and ownership friction

F1: Tech requirements - **Question:** "Which are must-haves?" - **Input:** Multi-select - **Options:** "Apple CarPlay", "Android Auto", "Driver-assist", "Heated seats", "Heat pump", "Premium audio" - **Helper:** "Must-haves are treated as filters; nice-to-haves affect ranking."

F2: Service tolerance - **Question:** "How far are you willing to travel for service?" - **Input:** Single-select - **Options:** "≤10 miles", "≤25 miles", "≤50 miles", "I'm flexible" - **Branching:** affects ownership friction if brand service footprint is limited (requires separate dataset).

Deep Fit results - Adds a "Your fit settings" expandable section showing: required range, buffers, charging persona, budget filters, and why certain models were filtered out. - Adds "Compare a specific EV" module with VIN/listing paste option.

Scoring model with formulas, weights, and buffers

Vehicle data schema needed (assumption)

OFFOLabs should store per trim (or per model-year where trim granularity isn't possible yet):

- `epa_range_mi` (combined) and `epa_mpge` or `kwh_per_100mi` from FuelEconomy.gov datasets. [21]
- `body_style`, `seats`, `cargo_class`, `awd`, `towing_capable`
- `msrp_estimate` or market price ranges (source not specified here)
- `dc_fast_charge_rating` (kW or 10–80% time) and `ac_charge_rate` (kW) — **not** from EPA dataset, typically OEM/third-party; present transparently as such.

Constraint filtering

A vehicle must pass all constraints to be eligible.

Range constraint

Define:

- `HardDayMiles` = max(longest_day_last_month, longest_day_typical_year × 0.85)

- Rationale: users often overestimate "typical year" hindsight; we temper with 0.85 to avoid over-filtering, but keep "hard day" dominant.

- `ComfortUseableFraction` = 1 − `MinArrivalSOC`

- `MinArrivalSOC` from Deep Fit (10%, 20%, 30%, 40% → 0.10–0.40)

- `ClimateMultiplier`

- Mild baseline: 1.00
- Hot: 1.10
- Cold: 1.25 (and up to 1.35 if "freezing most weeks" + "outdoors exposed")

- Justification: AAA reports large HVAC-related reductions at 20°F and more modest at 95°F; OFFO uses conservative buffers rather than claiming exact personal impact. [17]

- `LoadMultiplier`

- None: 1.00
- Roof/hitch rack often: 1.08
- Occasional towing/heavy loads: 1.15
- Frequent towing/heavy loads: 1.25

- Justification: AAA heavy cargo test shows ~24.5% reduction in a high payload condition in one example; OFFO uses 1.25 as a conservative "frequent load" case. [15]

- `ParkingMultiplier`

- Garage: 1.00
- Outdoors exposed: 1.05
- Rationale: battery/cabin conditioning and cold soak can increase friction; EPA methodology includes cold tests with HVAC settings, supporting buffer logic. [9]

Compute required rated range:

```
RequiredEPA = HardDayMiles / ComfortUseableFraction
RequiredEPA *= ClimateMultiplier * LoadMultiplier * ParkingMultiplier
RequiredEPA *= RoadTripMultiplier
```

Where `RoadTripMultiplier` is: - Rarely: 1.00
- Monthly: 1.05
- Weekly: 1.10
This is a "convenience" multiplier to avoid tight matches that make weekly road-trips painful.

Pass condition: `epa_range_mi >= RequiredEPA`

Charging feasibility constraint

Compute a household/work charging capability score:

- Estimate "miles added per hour" for AC charging:
- Level 1 is slow and can take very long to charge a BEV from empty to 80% per DOT; Level 2 is substantially faster. [22]

- For modeling simplicity:
  - L1 effective miles/hr: 3 (conservative placeholder; actual varies)
  - L2 effective miles/hr: 12 (conservative placeholder)
- If you implement more precise AC kW modeling later, replace these heuristics.

Compute `DailyChargeMiles = MilesPerHour × ChargingHoursAvailablePerDay`

- If user is home-reliable and `DailyChargeMiles >= WeekdayMiles`, pass.
- If user is public-dependent and "avoid fast charging," require either (a) work charging availability or (b) higher range margin (`RequiredEPA` + 10%).

Charger density fallback: if ZIP provided, compute a `ChargerDensityScore` using NREL API (public fast chargers within 5–10 miles and within a typical road-trip corridor), but treat it as an input to preference scoring, not a hard constraint. [23]

Budget constraint

- If user gives max price: `vehicle_price_est <= MaxPrice`
- If monthly payment: use a simple finance estimator (assumption) and filter by `EstimatedPayment <= TargetMonthly + tolerance`

Preference scoring

Eligible vehicles are scored 0–100 with weighted components.

Overall score:

```
TotalScore = 0.25 * ChargingFit
           + 0.25 * RangeFit
           + 0.15 * UtilityFit
           + 0.15 * CostFit
           + 0.10 * TechFit
           + 0.10 * OwnershipFriction
```

Weights reflect that charging and range are the primary satisfaction drivers; adjust by observing user outcomes.

Charging Fit (0–100)

Inputs: - `HomeFeasibility` (0–1) from dwelling + plug + dwell time
- `PublicFriction` (0–1) from patience and "avoid fast charging"
- `ChargerDensityScore` (0–1) from NREL charger count/fast availability near ZIP, when available. [24]

Example:

```
ChargingFit = 100 * (0.45*HomeFeasibility + 0.25*(1-PublicFriction) +
0.30*ChargerDensityScore)
```

If ZIP not provided, set `ChargerDensityScore = 0.5` and disclose this assumption.

Range Fit (0–100)

Compute margin:

`RangeMargin = (epa_range_mi - RequiredEPA) / RequiredEPA`

Clamp: - If RangeMargin < 0 → 0 (but this should have been filtered out) - If RangeMargin >= 0.30 → 100 - Else → `RangeFit = 100 * (RangeMargin / 0.30)`

Utility Fit (0–100)

- If body style matches: +40
- If seats match: +30
- If towing required and vehicle not towing-capable: 0 (or filtered)
- If user has heavy cargo needs: adjust by cargo class

Cost Fit (0–100)

- Price proximity: `CostFit_price = 100 * max(0, 1 - (Price / MaxPrice))` (scaled)
- Energy cost per mile:
- `kWh_per_mile = (kWh_per_100mi / 100)` from EPA dataset where available. [21]
- `EnergyCostPerMile = kWh_per_mile * $/kWh`
- $/kWh is user-entered or EIA state estimate. [20]

Combine: `CostFit = 0.7*CostFit_price + 0.3*CostFit_energy`

Tech Fit (0–100)

Treat "must haves" as either filters or penalties: - Must-have missing → 0 (or heavy penalty)
- Nice-to-have present → + points

Ownership Friction (0–100)

Inputs (all optional until you have the datasets): - service distance tolerance vs service footprint
- comfort with new brands
- maintenance considerations (if you later integrate reliability datasets)

In MVP, compute from: - "Service travel willingness" (higher willingness lowers friction) - "Public-dependent + avoid fast charging" increases friction

Example calculations for three archetypes

These are **illustrative** using hypothetical vehicles (A/B/C) to show mechanics. They are not claims about real models.

Hypothetical vehicle catalog

Vehicle	Type	EPA range (mi)	Price	kWh/100mi	Notes
A	Compact crossover	260	$38k	28	Balanced
B	Long-range sedan	330	$48k	26	Strong range
C	Electric truck	310	$62k	45	Utility focus

EPA range/efficiency values should be sourced from FuelEconomy.gov datasets in production. [21]

Archetype 1: Home-reliable commuter

Inputs: - Home charging yes; dwell 12+ hours; mild climate; garage - Weekday 30 mi/day; longest day last month 120; rare long trips - MinArrivalSOC = 20% - Budget $50k; body style “Crossover”

Buffers: - ComfortUseableFraction = 0.80 - ClimateMultiplier 1.00, Load 1.00, Parking 1.00, RoadTrip 1.00 - HardDayMiles = 120

RequiredEPA = 120 / 0.80 = **150 mi**

All vehicles pass range. Budget filters out C (>$50k). Eligible: A, B.

RangeFit: - A margin = (260-150)/150 = 0.733 → capped at 100 - B margin = (330-150)/150 = 1.2 → capped at 100

ChargingFit: high for both; minimal difference (user is home reliable).

UtilityFit: - A matches crossover → high - B sedan mismatch → lower

TotalScore (example): - A: Charging 95, Range 100, Utility 90, Cost 70, Tech 60, Friction 85
Total ≈ 0.25*95 + 0.25*100 + 0.15*90 + 0.15*70 + 0.10*60 + 0.10*85 = **88.0** - B: Charging 95, Range 100, Utility 55, Cost 50, Tech 60, Friction 85
Total ≈ **76.8**

Rank: **A > B** (recommend crossover first, sedan as alternative).

Archetype 2: Public-dependent road-tripper

Inputs: - No home charging; public charging weekly; “want fewer/shorter stops” - Longest day typical year 350; monthly long trips; highway 70% - Hot climate; outdoor parking - MinArrivalSOC 30% (wants buffer) - Budget $55k; body style “Any”

Buffers: - ComfortUseableFraction = 0.70 - ClimateMultiplier = 1.10 (hot) - ParkingMultiplier = 1.05 (outdoors) - RoadTripMultiplier = 1.05 (monthly)

HardDayMiles: max(last_month unknown, 350*0.85=298) → use 298

RequiredEPA = 298 / 0.70 = 425.7
× 1.10 × 1.05 × 1.05 ≈ **514 mi**

Result: **No vehicles pass** with this strict comfort buffer—this is expected and valuable feedback. The product should respond by: - explaining which assumptions are driving the requirement (high comfort SOC + public dependence + hot/outdoor + long days),
- allowing the user to adjust: “Would you accept arriving with 20% battery on road trips?” or “Would you consider a mid-trip fast charge?”

If user relaxes MinArrivalSOC to 20% and accepts 1 fast charge stop, OFFO can model “trip with one mid-charge” by using corridor charger availability via NREL “nearby route,” turning range into a “leg length” requirement rather than full-day. [25]

This is exactly the kind of honesty that increases trust.

Archetype 3: Towing/outdoor hobbyist

Inputs: - Home charging yes; cold climate; outdoors exposed - Frequent towing/heavy loads - Longest day last month 180; weekly long days; MinArrivalSOC 20% - Budget $70k; wants “Truck / towing”

Buffers: - ComfortUseableFraction 0.80 - ClimateMultiplier 1.35 (cold + outdoors + freezing most weeks) - LoadMultiplier 1.25 (frequent heavy loads; AAA payload evidence supports large reductions) [15]
- ParkingMultiplier 1.05, RoadTripMultiplier 1.10

HardDayMiles 180

RequiredEPA = 180 / 0.80 = 225
× 1.35 × 1.25 × 1.05 × 1.10 ≈ **437 mi**

Again, no vehicle passes; this reflects reality: towing + cold + high buffer is hard. OFFO should: - offer alternatives (PHEV suggestion if your product includes it),
- or recommend “larger pack + planned charging,”
- and explicitly cite that cold HVAC impacts and loads can reduce range materially, so buffers are warranted. [26]

Transparent recommendation templates and limitations

Vehicle recommendation explanation template

Use this as the "Why this EV fits you" block for each result.

Title: "Why we recommended this EV for you"

Fit summary (short): - "Great match for your **[charging setup]** and **[body style]**, with enough range buffer for your **[hard day]**."

Inputs we used (show user's answers): - Charging: "Home charging: Yes (Level 1 / Level 2 / unknown), Dwell time: 12+ hours" - Driving: "Longest day: 140 miles; Highway: 60%; Long-trip frequency: Monthly" - Comfort buffer: "Minimum arrival battery: 20%" - Climate: "ZIP-based climate category: Cold / Mild / Hot (or manual)" - Load: "Towing: Occasional / Frequent" - Budget: "Max price: $45,000" - Body style: "Crossover"

Data we used (show sources + timestamps): - "EPA rated range and efficiency from FuelEconomy.gov datasets (official label values). Data pulled: {timestamp}." [8]
- "Recall status from NHTSA recall data (when available for this make/model/year/VIN). Last checked: {timestamp}." [27]
- "Nearby public charging density from NREL Alternative Fuel Stations API (AFDC). Last checked: {timestamp}." [24]
- "Estimated home electricity rate: user-entered / EIA state average (source: EIA Electric Sales, Revenue, and Average Price). Snapshot date: {timestamp}." [20]

Range math (show numbers): - "Your required EPA range (with buffers): **{RequiredEPA} miles**" - "This vehicle's EPA range: **{epa_range_mi} miles**" - "Range margin: **{RangeMargin%}**"

Why buffers exist (cite and explain): - "Cold and HVAC can reduce real-world range; AAA testing reports large reductions at ~20°F with HVAC compared to ~75°F. We apply a conservative buffer rather than predicting an exact number." [14]
- "Loads/towing can reduce range; AAA payload testing showed significant range drop under heavy load in one scenario. We apply a load buffer when you tow or haul heavy gear." [15]
- "EPA's methodology includes hot and cold test cycles; real-world results still vary by speed, wind, elevation, and driving style." [9]

Limitations (explicit): - "Public charging availability changes over time."
- "EPA range is a standardized estimate; your real-world range may be higher or lower." [9]
- "Fast-charging speed can vary by trim, battery temperature, and station power; we show it when we have verified data."

"Not recommended" template

Title: "Why we didn't recommend this EV"

- "It fails your constraints:"

- “Required EPA range: {RequiredEPA} vs vehicle: {epa_range_mi}”
- or “Price {price} exceeds your max {maxPrice}”
- or “Your charging setup (no home/work charging) + ‘avoid fast charging’ makes it high-friction.”

Offer next action: - “Adjust your comfort buffer (min arrival battery)”
- “Consider allowing 1 fast-charge stop on long days”
- “Switch to Deep Fit for a more accurate route/charger model.”

Branching flow Mermaid diagram

```
flowchart TD
  A[Start Quick Fit] --> B{Primary charging?}
  B -->|Home| B1[Home access mini-check]
  B -->|Work| B2[Days/week charge at work]
  B -->|Public| B3[Fast-charge tolerance]
  B -->|Not sure| C

  B1 --> C[Weekly miles]
  B2 --> C
  B3 --> C
  C --> D[Longest day last month]
  D --> E[Long-trip frequency]
  E --> F[Budget]
  F --> G[Body style]
  G --> H{ZIP provided?}
  H -->|Yes| I[Quick Fit results + offer Deep Fit]
  H -->|No| H1[Manual climate]
  H1 --> I

  I --> J[Start Deep Fit]
  J --> K[Charging reality module]
  K --> L[Driving + comfort SOC]
  L --> M[Climate + parking exposure]
  M --> N[Utility + towing]
  N --> O[Budget + electricity rate]
  O --> P[Tech + service tolerance]
  P --> Q[Deep Fit results + explain math]
```

Analytics instrumentation mapping

Use a consistent schema: `flow_version`, `step_id`, and `branch_path` so you can cohort Quick Fit vs Deep Fit.

Question / step	Analytics event	Required properties
Start Quick Fit	`evfit_quick_start`	`flow_version`, `source`, `utm_campaign`, `device`, `geo_estimate`
Primary charging location	`evfit_charging_anchor_set`	`charging_anchor`, `confidence_level`, `branch_path`
Home access mini-check	`evfit_home_access_set`	`home_access`
Work days/week	`evfit_work_days_set`	`work_days`
Fast-charge tolerance	`evfit_public_tolerance_set`	`tolerance_level`
Weekly miles	`evfit_weekly_miles_set`	`weekly_miles`
Longest day last month	`evfit_hard_day_set`	`hard_day_miles`
Long-trip frequency	`evfit_long_trip_freq_set`	`freq`
Budget	`evfit_budget_set`	`budget_type`, `max_price`, `monthly_target`, `lease_buy`
Body style	`evfit_body_style_set`	`body_style`
ZIP entered / skipped	`evfit_zip_set`	`zip_provided`, `zip_prefix` (store prefix only), `manual_climate`
Quick Fit results viewed	`evfit_quick_results_view`	`num_recommended`, `top_vehicle_ids`, `constraint_binding`
Start Deep Fit	`evfit_deep_start`	`from_quick_results`, `branch_path`
Min arrival SOC	`evfit_min_soc_set`	`min_soc_pct`
Parking exposure	`evfit_parking_set`	`parking_type`
Towing/load	`evfit_load_set`	`towing_freq`, `rack_usage`, `weight_lbs_optional`
Electricity rate	`evfit_electricity_rate_set`	`rate_source` (user/eia), `rate_value`
Deep Fit results viewed	`evfit_deep_results_view`	`required_epa_range`, `top_vehicle_ids`, `filtered_count`

Question / step	Analytics event	Required properties
Recommendation explanation expanded	`evfit_explain_open`	`vehicle_id`, `section` (range/charging/cost/trust)
"Compare VIN/listing" used	`evfit_compare_initiated`	`input_type` (vin/listing), `vin_hashed`

Privacy and paywall UX copy snippets

ZIP collection copy

Inline text under ZIP field:
"ZIP helps us estimate climate effects and nearby public charging. We only use it to compute your fit and charger density. You can skip this and choose climate manually."

Data handling disclosure (expandable):
"We store ZIP at the ZIP-5 level only if you save your results. You can delete it anytime in Settings."

VIN collection copy (optional "compare a specific EV" feature)

Use VIN as an optional enrichment step; treat it as sensitive.

Inline text:
"VIN lets us identify the exact build details and check for open recalls. We can also work with just a make/model if you prefer."

Data handling disclosure:
"A VIN can be considered personally identifying because it uniquely identifies a vehicle. We store VIN only if you save it to your Garage, and you can delete it anytime." [18]

Delete/export controls: - "Delete saved vehicle"
- "Export my saved vehicles (CSV)"
- "Delete my EV Fit profile"

Paywall transparency copy (if premium exists)

Before any gated step:
"Next: Premium insights (optional). You'll see pricing before you pay—no surprises."

Pricing page header:
"What you get in Premium" + "Price" + "Cancel anytime"
If your product is free now, explicitly say: "This feature is currently free during beta."

A/B test plan for fit accuracy and satisfaction

What to validate

You have two truths to measure:

1) **Perceived fit quality**: "These recommendations make sense for me."
2) **Behavioral confirmation**: shortlist saves, return visits, dealer messages, and ultimately test-drive/appointment conversions (if connected to dealers later).

High-priority tests

Test	Hypothesis	Primary metric	Sample-size guidance	Timeline
Quick Fit vs Quick Fit + hard-day emphasis	Rewording "hard day" and showing its impact increases completion and trust	Completion rate; "fit confidence" (1–5)	If baseline completion 55% and want +5pp, expect ~1k–2k users total	2–3 weeks
Default MinArrivalSOC 20% vs 30%	Higher default reduces "surprise range anxiety" reports but may over-filter	"I'd consider these EVs" rate; % with "no matches"	Start with 50/50 split; watch no-match rate	3–4 weeks
ZIP optional framing	Stronger transparency increases ZIP entry without harming completion	ZIP entry rate; completion rate	~1k sessions	2–3 weeks
Deep Fit "Explain my math" toggle	Transparent math increases satisfaction and reduces churn	Explain-open rate; D7 retention; support contacts	~2k sessions	4–6 weeks

Notes on sample sizing: exact required N depends on baseline rates and desired detectable lift. Use a standard two-proportion power calculation with $\alpha=0.05$ and power=0.8; adjust for segmentation (device, geo) and for marketplace contexts where outcomes cluster by region.

Closing implementation notes

- Treat **range buffering as a conservative planning tool**, not a promise. Cite AAA and EPA methodology in the UI explanation so users understand why buffers exist. [17]
- When a user gets "no matches," don't dead-end—offer **guided relaxations** (lower min SOC, allow a mid-charge stop, consider PHEV if in scope) and show which assumption is binding.
- Make "data freshness" explicit for each dataset: EPA dataset pull date, NHTSA recall check timestamp, charger density timestamp from NREL/AFDC, and electricity price snapshot date from EIA. [28]

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